# KCPC Week 5 Reading
# Graphs II: Dijkstra and 0–1 BFS

Follow this guide alongside the slides and problem sheet. Each reference below covers specific foundational theory from the workshop: graph traversal (DFS/BFS), shortest paths with non-negative weights (Dijkstra), the relaxation invariant, priority queues, and 0–1 BFS. Page numbers refer to the *printed* pages of each text, rather than the PDF viewer counter. For the original research behind these algorithms, see *Week 5 Papers*.

## I. Core algorithmic reading

### Guide to Competitive Programming (Antti Laaksonen-style community PDF)

https://duoblogger.github.io/assets/pdf/memonvyftw/guide-t-cp.pdf

- **Depth-First Search, p. 83:** stack-based exploration and reachability; matches the slides' graph-foundations section.
- **Breadth-First Search, p. 85:** FIFO layers and why first discovery minimises edge count; needed for Problem 7 (01 Matrix).
- **Dijkstra's Algorithm, p. 89:** relaxation with a priority queue for non-negative weights; the basis for Problems 1–2, 4–5 and 9–15.
- **Bellman–Ford, p. 88 (further reading):** the negative-weight alternative the slides warn about; not required for this sheet.
- **Floyd–Warshall, p. 92 (further reading):** all-pairs shortest paths; optional context for why we only need single-source methods here.

## II. 0–1 BFS

### CP-Algorithms

https://cp-algorithms.com/graph/01_bfs.html

A concise explanation of 0–1 BFS, including its correctness argument and time complexity analysis. Read it before attempting the deque-based problems: 3 (arrow grid), 6 (obstacle removal), 8 (Wizard in Maze) and 15 (Stronger Takahashi). Compare its dial/bucket viewpoint with the slides' "two adjacent distance levels" invariant, and with Dial (1969) in *Week 5 Papers*.

## III. Formal textbook reference

### Introduction to Algorithms (Cormen, Leiserson, Rivest, Stein)

https://www.cs.mcgill.ca/~akroit/math/compsci/Cormen%20Introduction%20to%20Algorithms.pdf

- **Priority queues, p. 162 (§6.5):** the heap operations behind the slides' lazy min-heap implementation.
- **Breadth-First Search, p. 594 (§22.2):** the formal layering argument; also covers the shortest-paths-in-unweighted-graphs theorem.
- **Depth-First Search, p. 603 (§22.3):** reachability and edge classification.

- **Dijkstra's Algorithm, p. 658 (§24.3):** the formal proof of the settling invariant the slides rely on, including why negative weights break it.
- **Bellman–Ford, p. 651 (§24.1, further reading):** the correct tool when negative weights appear.
- **Floyd–Warshall, p. 693 (§25.2, further reading):** optional all-pairs perspective.

## Suggested order

Start with the competitive programming guide (DFS p. 83, BFS p. 85, Dijkstra p. 89) for implementation clarity, alongside the slides' worked traces; solve Problems 1–3. Read the CP-Algorithms 0–1 BFS page, then attempt Problems 3, 6, 8 and 15. Use CLRS §§22.2–22.3 and §24.3 for the formal invariants and complexity analysis when writing up your own reasoning, and consult Bellman–Ford and Floyd–Warshall only as further reading. For original sources and historical context, see *Week 5 Papers*.